

Structured Query Language (SQL)

Comprehensive Theory, Command Reference & 25 Practice Questions

1. Introduction & SQL Theory

SQL (Structured Query Language) is the standard domain-specific language used to manage, manipulate, and retrieve data stored in **Relational Database Management Systems (RDBMS)** such as MySQL, PostgreSQL, Oracle, SQL Server, and SQLite.

Key Database Concepts

- **Database:** An organized collection of structured data stored electronically.
- **Table:** A collection of related data entries consisting of columns (fields/attributes) and rows (records/tuples).
- **Primary Key:** A column (or set of columns) that uniquely identifies each row in a table. It cannot contain NULL values.
- **Foreign Key:** A field in one table that uniquely identifies a row in another table, creating a relationship between them.
- **NULL Value:** Represents missing, unknown, or inapplicable data. It is different from zero or a blank space.

SQL Execution Order

Although SQL queries are written starting with SELECT, the RDBMS processes clauses in the following sequence:

FROM / JOIN → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → LIMIT / OFFSET

2. SQL Command Categories & Examples

SQL commands are categorized into five primary sub-languages based on their functionality:

Category	Description	Commands	Practical Example
DDL Data Definition Language	Defines and modifies the physical structure of database objects (tables, schemas).	CREATE, ALTER, DROP, TRUNCATE, RENAME	<code>CREATE TABLE Employees (EmpID INT PRIMARY KEY, Name VARCHAR(50), Salary DECIMAL(10,2));</code>
DML Data Manipulation Language	Manages and modifies the data present inside the existing tables.	INSERT, UPDATE, DELETE	<code>INSERT INTO Employees VALUES (101, 'Alice', 75000);</code> <code>UPDATE Employees SET Salary = 80000 WHERE EmpID = 101;</code>
DQL Data Query Language	Used to retrieve data from the database based on specified criteria.	SELECT	<code>SELECT Name, Salary FROM Employees WHERE Salary > 50000 ORDER BY Salary DESC;</code>

Category	Description	Commands	Practical Example
DCL Data Control Language	Controls permissions, access levels, and rights on database objects.	GRANT, REVOKE	<code>GRANT SELECT, INSERT ON Employees TO user_john;</code>
TCL Transaction Control Language	Manages transactions to maintain database integrity and consistency.	COMMIT, ROLLBACK, SAVEPOINT	<code>BEGIN TRANSACTION; DELETE FROM Employees WHERE EmpID = 101; ROLLBACK;</code>

3. Important SQL Operations

A. SQL Joins

Joins combine rows from two or more tables based on a related column between them.

- **INNER JOIN:** Returns records that have matching values in both tables.
- **LEFT (OUTER) JOIN:** Returns all records from the left table, and matched records from the right table.
- **RIGHT (OUTER) JOIN:** Returns all records from the right table, and matched records from the left table.
- **FULL (OUTER) JOIN:** Returns all records when there is a match in either left or right table.

```
SELECT Orders.OrderID, Customers.CustomerName
FROM Orders
INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID;
```

B. Aggregate Functions & GROUP BY

Aggregate functions perform a calculation on a set of values and return a single value: COUNT(), SUM(), AVG(), MIN(), MAX().

```
SELECT Department, AVG(Salary) AS AvgSalary
FROM Employees
GROUP BY Department
HAVING AVG(Salary) > 60000;
```

4. SQL Practice Questions (25 MCQs)

1. What does SQL stand for?

- A) Strong Question Language
- B) Structured Query Language
- C) Sequential Query Language
- D) Standard Question List

2. Which SQL command category does the 'CREATE' statement belong to?

- A) DML
- B) DQL
- C) DDL
- D) TCL

3. Which clause is used to filter records in SQL before grouping?

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|-------------|-------------|
| A) HAVING | B) WHERE |
| C) ORDER BY | D) GROUP BY |

4. Which command is used to permanent save transactions in SQL?

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|--------------|-------------|
| A) SAVEPOINT | B) ROLLBACK |
| C) COMMIT | D) GRANT |

5. How do you select all columns from a table named "Customers"?

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|----------------------------|-----------------------------|
| A) SELECT Customers; | B) SELECT * FROM Customers; |
| C) GET ALL FROM Customers; | D) SELECT ALL Customers; |

6. Which SQL statement is used to update data in a database?

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| A) SAVE | B) MODIFY |
| C) UPDATE | D) ALTER |

7. Which clause is used with aggregate functions to filter group results?

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| A) WHERE | B) HAVING |
| C) LIKE | D) ORDER BY |

8. Which command removes all records from a table without logging individual row deletions?

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|-------------|-----------|
| A) DELETE | B) DROP |
| C) TRUNCATE | D) REMOVE |

9. What constraint ensures that all values in a column are distinct?

- | | |
|-------------|------------|
| A) NOT NULL | B) UNIQUE |
| C) CHECK | D) DEFAULT |

10. Which SQL join returns all rows from the left table, even if there are no matches in the right table?

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|---------------|---------------|
| A) INNER JOIN | B) RIGHT JOIN |
| C) LEFT JOIN | D) FULL JOIN |

11. Which wildcard character in SQL represents zero or more characters?

- | | |
|-------------------|----------------|
| A) _ (underscore) | B) % (percent) |
| C) * (asterisk) | D) # (hash) |

12. What keyword is used to return only distinct (different) values?

- | | |
|--------------|-----------|
| A) DIFFERENT | B) UNIQUE |
| C) DISTINCT | D) MATCH |

13. Which default order does ORDER BY sort the results in?

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|---------------|----------------|
| A) Descending | B) Ascending |
| C) Random | D) Index-based |

14. Which statement is used to remove a table structure permanently from the database?

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|-------------------|-----------------|
| A) DELETE TABLE | B) DROP TABLE |
| C) TRUNCATE TABLE | D) REMOVE TABLE |

15. Which function is used to count the number of rows in a SQL table?

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|------------|-------------|
| A) SUM() | B) TOTAL() |
| C) COUNT() | D) NUMBER() |

16. Which of the following is NOT a TCL command?

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|-----------|--------------|
| A) COMMIT | B) ROLLBACK |
| C) GRANT | D) SAVEPOINT |

17. How do you test for NULL values in SQL?

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|------------|----------------|
| A) = NULL | B) IS NULL |
| C) == NULL | D) EQUALS NULL |

18. Which operator is used to search for a specified pattern in a column?

- | | |
|------------|-----------|
| A) IN | B) LIKE |
| C) BETWEEN | D) EXISTS |

19. What SQL command adds a new column to an existing table?

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|---------------------------|-------------------------|
| A) ALTER TABLE ... ADD | B) UPDATE TABLE ... ADD |
| C) INSERT INTO ... COLUMN | D) MODIFY TABLE ... ADD |

20. What is the primary purpose of a Foreign Key?

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|-------------------------------|-----------------------------|
| A) Uniquely identify a record | B) Link two tables together |
| C) Speed up search queries | D) Encrypt sensitive data |

21. Which statement combine the result-set of two or more SELECT statements (excluding duplicates)?

- A) JOIN
- B) UNION
- C) INTERSECT
- D) MERGE

22. Which command is used to revoke privileges from a database user?

- A) REMOVE
- B) REVOKE
- C) DENY
- D) CANCEL

23. Which clause limits the number of rows returned in MySQL/PostgreSQL?

- A) TOP
- B) LIMIT
- C) ROWNUM
- D) MAX

24. What type of function is AVG()?

- A) Scalar Function
- B) Aggregate Function
- C) String Function
- D) System Function

25. Which order of SQL execution is correct?

- A) SELECT -> FROM -> WHERE
- B) FROM -> WHERE -> SELECT
- C) WHERE -> FROM -> SELECT
- D) SELECT -> WHERE -> FROM

Answer Key

- | | | | | |
|------|-------|-------|-------|-------|
| 1. B | 6. C | 11. B | 16. C | 21. B |
| 2. C | 7. B | 12. C | 17. B | 22. B |
| 3. B | 8. C | 13. B | 18. B | 23. B |
| 4. C | 9. B | 14. B | 19. A | 24. B |
| 5. B | 10. C | 15. C | 20. B | 25. B |